

SOFTWARE ENGINEERING

SYNOPSIS / PROJECT PROPOSAL

Voice Assistant for Multimedia Controls

Group No.: 12

1. Project Title

Voice Assistant for Multimedia Controls

2. Group Number

12

3. Team Profile

Sl. No.	Name	SRN
1	Prajwal GT	PES2UG25CS818
2	Shreedhar R	PES2UG25CS823
3	Shrinivas S J	PES2UG24CS494
4	Sarang S Nair	PES2UG24CS447

Note: Every team member will be involved in all major project activities, including requirements analysis, design discussions, implementation, integration, testing, documentation and demonstration.

4. Proposed Project Description

The proposed project is a brand-new hardware-based Voice Assistant for Multimedia Controls. The system is designed to provide hands-free control of common multimedia operations through simple, predefined voice commands. A Raspberry Pi or ESP-based controller, microphone and Python software will be used to capture and process voice input. After a supported command is recognized, the system will communicate the corresponding multimedia action to a connected computer or media device.

The core system will support volume up, volume down, play, resume, pause, next and previous operations. MediaPipe may also be integrated for predefined hand-gesture recognition and gesture-to-command mapping. This provides an alternative hands-free interaction method when speaking is inconvenient. The system will be designed as a modular prototype so that additional commands, gestures and multimedia devices can be added later.

The usual users of the system are students and general multimedia users who frequently use music, video or presentation applications. The project does not currently depend on a specific external customer; it is proposed as an academic prototype intended for demonstration and evaluation.

4.1 Usual Users

- Students using computers for music, videos and presentations.
- Users who prefer hands-free multimedia control.
- Users who want to control media from a short distance.
- Users looking for an alternative to keyboard, mouse or touchscreen controls.

4.2 Functional Features

- Recognize predefined voice commands.
- Increase media volume using a voice command.
- Decrease media volume using a voice command.
- Play or resume currently paused media.

- Pause currently playing media.
- Move to the next media track or video.
- Move to the previous media track or video.
- Provide feedback when a supported command is recognized.
- Ignore unsupported or unrecognized commands safely.
- Support selected gesture-based multimedia controls using MediaPipe, if implemented.

5. Plan of Work and Product Ownership

The following plan covers the short-term work planned for the next few weeks and identifies the primary product ownership of each member.

5.1 Plan of Work

Week	Planned Work
1	Finalize requirements; identify hardware/software requirements; study voice recognition and multimedia control methods.
2	Set up Raspberry Pi/ESP, microphone and communication with the target computer; develop the basic voice-input module.
3	Implement play, pause, resume, volume-up and volume-down commands.
4	Implement next and previous controls and integrate multimedia commands.
5	Implement selected MediaPipe gesture recognition and gesture-to-command mapping.
6	Integrate all modules; perform testing; fix errors; improve response time and command/gesture recognition accuracy.
7	Perform system-level testing, final demonstration, documentation and feedback.

5.2 Who Should Do What – Product Ownership

Team Member	SRN	Functional Features Owned	Qualitative Contribution
Prajwal GT	PES2UG25CS818	Voice command detection; play/resume; pause	Improve voice-command recognition accuracy and reduce response delay.
Shreedhar R	PES2UG25CS823	Volume up; volume down; Raspberry Pi/ESP and device communication	Ensure reliable hardware-to-device communication and consistent volume control.
Shrinivas S J	PES2UG24CS494	Next; previous; command feedback	Ensure reliable track navigation and clear feedback when commands are recognized.
Sarang S Nair	PES2UG24CS447	MediaPipe gesture recognition; gesture-to-command mapping; system integration	Improve gesture recognition accuracy and reduce unintended gesture triggers.

Product ownership is divided by primary functional features, but all four members will participate in every major project activity. The ownership assignments ensure that each member has a clearly defined contribution to the final product.

5.3 Qualitative Project Goals

- Performance: recognized commands should be executed with minimal noticeable delay.
- Accuracy: supported commands and configured gestures should be recognized correctly under normal conditions.
- Reliability: the system should operate continuously without frequent crashes or interruptions.
- Usability: commands and gestures should be simple, consistent and easy to understand.

- Maintainability: the software should be modular so that new commands can be added easily.
- Portability: the prototype should run on suitable Raspberry Pi or ESP-based hardware.
- Robustness: invalid commands, background noise and temporary communication failures should be handled gracefully.
- Privacy: voice input should not be unnecessarily stored or transmitted externally.

6. Expected Outcome

The expected outcome is a working prototype that allows users to control common multimedia operations through voice commands, with optional gesture-based control using MediaPipe. The project will demonstrate the application of software engineering principles to a practical hardware-software system involving voice recognition, embedded hardware interfacing, computer vision and multimedia control.

7. Evaluation Readiness

The proposal contains the required project title and group number, complete team profile, detailed project description, target users, functional features, short-term plan of work, individual product ownership, functional responsibilities and qualitative contributions. The final submission can be provided as a single PDF document for Group 12.